

## Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

### (Lab 10)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

\* May use `main()` function approach (or multiple-functions) learnt in our earlier lesson in this lab

**Task:** build a program (for handling circles), with Procedural and Object-Oriented (OO) approaches:

- Given the radius of a circle, we need to calculate the *circumference* and *area* of the *circle*.
- Define a simple class of ICP students

1. Write a Python program (named `Lab10_1.py`) to generate mark 6 random numbers, using set:
  - Start with empty set, add elements within proper range of mark 6 numbers (1 to 49), until the size of the set is 6. Use function calling `random.randint(1,49)`
  - Show all random numbers picked (even duplicated)
  - Show the final result (the set with 6 *unique* numbers)

*Sample Program Output (may have **duplicated** number during the looping):*

```
Random picking starts...
27
42
11
28
11
20
31
The final set of Mark 6 is: {42, 11, 20, 27, 28, 31}
```

2. Write a Python program (named `Lab10_2.py`) to hold the information pairs of integer (1 to 10) to its cube value, with dictionary:
  - Use the integer as the key of dictionary and the value of dictionary is the cube, as `num:cube`
  - Display the whole dictionary by simple `print()` function

*Sample Program Output:*

```
The dictionary of num:cube pair (integer numbers 1 to 10) is:
{1: 1, 2: 8, 3: 27, 4: 64, 5: 125, 6: 216, 7: 343, 8: 512, 9: 729,
10: 1000}
```

~ END ~